KGuitar: Master Game Plan & Step-by-Step Implementation Roadmap

Owner Reference Edition (Post-11/9 Handoff Baseline) · Target Hardware: Samsung Galaxy Z Fold 8, Galaxy Tab S10+, Sonicake Pocket Master (USB-C), M-Vave Chocolate Plus (BLE MIDI), & MacBook Pro Audio Workstation · Branch Rule: Work strictly on `big-update` (never `main` until "ship it").

> **Golden Engineering Mandate ("Zero Corners Cut"):** Every stage in this master roadmap is designed for a private, single-user flagship build. Never substitute simplified fallbacks or toy algorithms. Always use the best available DSP, neural models, and studio sample libraries; keep paid-plugin sound packs in private phone/Google Drive storage (never committed to the public GitHub repo); and require a Co-Architect review before writing code for each stage.

0. Baseline Inventory: What Is Already Built (As of Commit 3b01f79)

Before executing Step 1, it is essential to recognize what is already completed and checkpointed on the `big-update` branch so Claude upgrades existing modules rather than rebuilding them from scratch:

- **Streamlined Top Menu & Song Player:** Top navigation is consolidated into `Songs · Learn · Jam · Tools · Ask · Settings`. `Play` is merged directly into `Songs` (`src/screens/PlayScreen.tsx`), featuring a persistent `NowPlaying.tsx` bar and `Settings` with `Cards` + `Manual`.
- **Yousician-Style Game Mode & Bouncing Ball:** Horizontal 6-string highway (`src/components/GameHighway.tsx`) with high `e` on top, color-coded left-hand finger blocks (1=Index to 4=Pinky), parabolic bouncing rhythm ball, and `Hear it` driving the highway (`TabPlayer.position()`).
- **9 Floating Cards & Resizable Tidy-Grid Windows:** Hovering cards (`src/float/`, `FloatCard.tsx`) for *Hand, Chord now & next, Mini neck, What you played, Bar & beat, Tempo, Quick tuner, Mini video,* and *Timer*, plus width/height resizable windows (`Win.tsx`).
- **Exercises & Lick Chainer:** Moved to `Learn → Exercises` (`src/exercises/ExercisesPage.tsx`) with *"For you"* and *"Make your own"* sections.
- **5-Layer Fretboard Geometry, Tunings & Core Styles:** Stackable CAGED + Triad X-Ray + Pentatonic + 3NPS Modal Bridge + Free-Finger Chord Embellishments, Voice-Leading arrows, Gold Target Tones, Taro Patch / Open G (`D-G-D-G-B-D`), Wahine, Mauna Loa, Drop D, Guitar, 20" Guitalele, Tenor Uke, Low-G Baritone Uke, and style paths shifted to Melodic Prog, Prog-Pop, Ambient Tapping, and Dylan Gossett Acoustic Singer-Songwriter.

Master Execution Order Overview (What to Do First, Next & After)

Stage	Core Focus Area	Why It Sits at This Exact Spot in the Order
Stage 1	**11/9 Handoff Test & Splitter Fixes**	Clears the 2 unfinished browser test items where Claude paused (`build25-test.cjs` & `windows-test.cjs`) so the baseline is 100% green before new code is added.
Stage 2	**Streamlined UI Polish, Floating Card Hooks & "Band Behind Me"**	Finishes Claude's queued medium tasks (simplifying `Jam` to 3 choices, publishing `Jam` state to Floating Cards, Glowing Tips, Songsterr/GP Band Behind Me + Moises 5-Fader Stem Mixer, Unified Songs Library, and PDF Manual).
Stage 3	**The "$25,000 Studio-Grade" Audio Engine & Mac Plugin Harvest**	**The Foundation Baseline:** Upgrading the audio engine and harvesting your Mac plugins *before* building Strum Machine, Riff Scanner, and Beat Maker ensures every feature immediately plugs into a 6-bus studio console and real multi-sampled instruments instead of synthetic math waves.
Stage 4	**Riff Scanner "Three-Sense" Overhaul & Pro Expansions**	Uses Stage 3's continuous MPE pitch-contour spine and Jam Origin benchmarks to solve every visual and audio transcription bottleneck, plus new pro computer-vision and scrubbers.
Stage 5	**Carbon-Copy "Strum Machine" + Multi-Genre Rhythm Engine**	Drops into `Jam → Play with a band` and `Play` on top of Stage 3's Acoustic Tri-Engine (Orange Tree + Ample SJ + Nembrini) and Electric Tri-Engine (Sampled DI → 8 Neural DSP Archetypes).
Stage 6	**Beat Maker Groovebox, "Match a Song's Vibe" & "Guitarcatcher"**	Starts with a visual mockup for approval, then builds the 4-Quadrant Full-Screen Cockpit, song-vibe generator, live MPE Guitar-to-Synth, VocalEssential/Humanoid chops, and MPC 16 Levels.
Stage 7	**"Pickup Music" Deep Mastery Course & Native Phone Stompbox**	Rebuilds `Learn` with 10 deep-dive units backed by Strum Machine and artist tones, and adds sub-8ms native Android `Oboe` + `MidiPlugin.java` live multi-FX pedalboard processing.

STAGE 1: Finish the 2 Active Test & Layout Fixes (From 11/9 Handoff)

Before adding new features, have Claude finish the exact three items listed under *"In progress when Claude stopped"* in `notes/GEMINI-HANDOFF.md`:

1. **Fix the Lick Chainer Browser Test (`tests/browser/build25-test.cjs` line ~272):** Because the Lick Chainer now lives inside `Learn → Exercises → Make your own`, update the test so after calling `sub(page, 'Exercises')`, it clicks `[data-testid=ex-sections] button:has-text("Make your own")` before waiting for `.chain-pick-add`.
2. **Fix the Phone Window Splitter Drag Check (`tests/browser/windows-test.cjs`):** Diagnose why *"Play phone: dragging the bar under Scales & chords did not make it taller"* failed on the phone viewport (`412×915`). Check whether the new persistent `NowPlaying` bar or a floating card overlaps the bottom resize handle on narrow screens, and fix the underlying layout/z-index cause without weakening the test.
3. **Full Multi-Viewport & Theme Verification:** Run `npm test` (logic) and all browser test suites across Fold cover (`412×915`), Fold inner (`840×757`), and Tablet (`1280×800`) in Day, Night, and Psychedelic themes with zero horizontal overflow.

STAGE 2: Streamlined Navigation, Floating Card Hooks & "Band Behind Me"

This stage completes Section B of your master plan (`notes/GEMINI-HANDOFF.md` items #4, #5, and #7) so the entire app's navigation is unified:

1. **Simplify `Jam` to 3 Choices & Connect Floating Cards:**
   - Present three clear top-level modes in `Jam`: **1. Play with a band**, **2. Make a beat**, and **3. Loop myself**.
   - Wire `Jam` to broadcast its live chord changes, upcoming chord, key, bar/beat, and BPM via `publishFloat('jam', {...})` (matching `PlayScreen.tsx`) so all 9 Floating Cards (*Chord now & next, Mini neck, Hand, Bar & beat, Tempo*) update live while jamming.
2. **Glowing Tips & Guitar Advice on the `Ask` Bubble:**
   - Display a subtle glowing indicator dot on the `Ask` bubble whenever a context-aware tip exists for the active screen, plus an "**Always show tips**" option.
   - Serve both **App Feature Tips** (from `src/learn/guide.ts`) and **Guitar Playing & Technique Advice**, with an independent toggle switch in `Settings`.
3. **"Band Behind Me" (`src/audio/band.ts`) + Moises-Style 5-Fader Stem Mixer in `Play`:**
   - **Multi-Track Songsterr / Guitar Pro Backing:** When playing a Songsterr or Guitar Pro tab in `Play`, render the companion band tracks (drums, bass, second guitar, keys/pads) in sync behind you via `src/audio/band.ts`.
   - **Automatic Chord Band Fallback:** When a tab only contains single-note melody or chord symbols without multi-track backing, automatically generate a locked rhythm section (drums + bass + rhythm accompaniment) from the tab's chord changes and style.
   - **Moises Pro-Style 5-Fader Stem Mixer:** Whenever 6-stem separation (`htdemucs_6s`) runs on a YouTube video or song file, persist all separated stems (*Guitar, Bass, Drums, Vocals, Other*) in IndexedDB and expose a 5-fader mixer in `Play` and `Game Mode`. You can mute the original guitar to `0%` (so you *are* the guitar), set it to `15%` as a quiet guide track while learning a hard bar, or solo the original guitar to study nuances.

4. **Unified `Songs` Library, One-Box "Add a Song", Default Shorts Channels & Illustrated PDF Manual:**
    - Merge song lists into **one unified `Songs` library** with instant filter pills (*All, Saved, Practice Plan, Scanned, Backing Tracks, By Style/Tuning*).
    - Replace separate import boxes with **one smart "Add a Song" input box** that auto-detects whether you pasted a YouTube/Shorts URL, typed a song/artist name for Songsterr search, dropped a `.gp/.gp5/.gpx`/audio/video file, or uploaded tab screenshots.
    - Pre-configure **Mr. Tabs**, **Level Up Guitar Tabs**, and **Kenneth Acoustic** as default curated channels in the Shorts scanner.
    - Build a 1-tap **Export Manual as PDF** feature, plus a **2-Minute Quick-Start Tour** and an **In-Depth Illustrated Master Manual** using automated Playwright Day-Mode screenshots explaining *How*, *Why*, and *When* to use every feature.

STAGE 3: The "$25,000 Studio-Grade" Audio Engine & Zero-Touch Mac Harvest

Why Building the Audio Engine Here Is the Baseline for Everything Else

Currently, `src/audio/engine.ts` is only 73 lines of code: every sound in the app dumps into a single `master` gain node, passes through one generic WebAudio compressor and soft-clipper, and exits through `finalIn → fold → ctx.destination`. Meanwhile, `src/audio/input.ts` copies `Float32Array` audio chunks over `postMessage` into a JavaScript array every block, `tools/guitar_di.py` feeds synthetic Python math waveforms into your amp captures (causing distorted chords to sound like a keyboard), and `src/audio/pluck.ts` plays acoustic chords by firing 6 isolated single-note plucks spaced 14–18ms apart (creating phase comb-filtering). Upgrading the primary audio engine and harvesting your Mac plugins **first** guarantees that Strum Machine, Band Behind Me, Beat Maker, Looper, and Riff Scanner are built directly on top of a unified, studio-grade foundation.

Part 3A: The 6-Bus Studio Summing Console & Real-Time DSP Spine

Studio Bus / Module	DSP Chain, Routing & Musical Purpose
Bus 1: Drums & Percussion	Transient punch shaper, parallel drum bus saturation, and a pre-fader kick trigger output feeding the sidechain duckers on Bus 2 and Bus 4.
Bus 2: Bass & Sub (Upright / Electric / 808)	28 Hz 24dB/oct subsonic rumble filter, mono low-end lock below 110 Hz, and phase-aligned kick-pocket sidechain ducking so the kick and bass never fight for headroom.
Bus 3: Guitars & Strum Machine Rhythm	Stereo acoustic/electric imaging, Nembrini acoustic body IR convolution send, and "**Auto-Pocket Lead Carve**": a dynamic sidechain EQ that automatically dips `1.5–3.2 kHz` by `-2.5 dB` in the backing guitars *only during the milliseconds you play a note on your real guitar*, letting your live lead sit effortlessly on top of the mix.
Bus 4: Keys, Synths & Vocal Chops	Musical "**Kick Pump**" sidechain envelope ducking, analog-modeled tape warmth filter, and mid-side stereo widener for Humanoid choirs, Rabea synth pads, andTim Henson Multivoicer chords.

Studio Bus / Module	DSP Chain, Routing & Musical Purpose
Bus 5: 4-Track Pro Looper & Perform FX	WASM SIMD `SignalsmithStretch` real-time pitch/time engine, zero-latency `-12st` Bass Octave DSP, 50% half-speed pitch-lock, and the tactile Perform FX strip with 1-tap internal resampling.
Bus 6: Live Guitar Input & "Guitarcatcher" Synth	Routes live USB-C (Sonicake Pocket Master) or mic input and the MPE guitar-to-synth engine into `finalIn` right before `fold`, preserving your mono/stereo hardware routing rules in `src/audio/rig.ts`.
3 Shared Studio Aux Sends ("Same-Room" Glue)	**Aux A (Studio Wood Room IR):** Binds Strum Machine's acoustic guitar, upright bass, and drums into one physical room. **Aux B (Lush Plate / Shimmer Hall):** For lead guitar and choir pads. **Aux C (Tempo-Synced Stereo Tape Delay):** Locked to the Master Clock.
Master Bus Glue & True-Peak Limiter	All 6 buses sum into a sidechain high-passed (90 Hz) SSL-style VCA bus glue compressor (2:1 ratio, slow 30ms attack, auto-release) followed by a 4x oversampled look-ahead brickwall limiter at `-0.3 dBFS` before `finalIn → fold`.

Part 3B: 6-String Physical Voice-Stealing, Zero-Jitter Memory & Master Clock

- **6-String Physical Voice-Stealing (`src/audio/pluck.ts`):** Model 6 physical string lanes (Strings 1–6). Plucking a new note on String 5 applies a realistic `12ms` finger-lift damping envelope to the previous note ringing on String 5 (instead of letting 12 overlapping notes ring into mud), while sympathetically resonating unplucked open strings.
- **Lock-Free `SharedArrayBuffer` Ring Buffers + "✨ Capture Last Pass" (`src/audio/input.ts`, `looper.ts`):** Replace main-thread `postMessage` chunk-copying with a lock-free `SharedArrayBuffer` + `Atomics` ring buffer (with automatic `ArrayBuffer` fallback). Maintain a continuous **3-minute rolling audio history** AND a **4-bar retroactive unrecorded note/pad buffer** so tapping "✨ **Capture Last Pass**" immediately commits a great riff or beat you just played even when `Record` was OFF.
- **Single Sample-Accurate Master Clock:** Drive `Jam`, Strum Machine, `Play`, `Game Mode`, the 4-Track Looper, and the Beat Maker from one unified `AudioWorklet` sample-counter clock so switching screens or looping A/B sections never drifts by a single millisecond.
- **Continuous MPE Pitch-Contour Spine (`src/audio/noteDetect.ts`):** Stream continuous fractional pitch (`midiFloat`), semitone bend offset, RMS amplitude envelope, and high-frequency pick-attack spectral flux (`onsetFlux`) at ~86–100 Hz to power Riff Scanner articulations, Game Mode bend arches, and live Guitarcatcher synth tracking simultaneously.

Part 3C: Zero-Touch MacBook Audio & Plugin Harvest (`tools/make-pro-banks.py`)

Using terminal access on your MacBook, Claude will scan `/Library/Audio/Plug-Ins/`, Kontakt paths, and `~/Downloads`, and use headless Python VST3/AU hosting (Spotify's `pedalboard` / `dawdreamer`) plus automated AppleScript/Logic Pro bouncing so you never have to manually configure tracks in Logic Pro. All rendered multi-take packs are stored in your local phone / Google Drive folder:

1. **Batch 1 — The Acoustic & Slack Key Tri-Engine:**
   - `acoustic` **Bank (Orange Tree Evolution Acoustic + Nembrini Acoustic Voice Pro):** Route Orange Tree through Nembrini's *Martin HD-28V / 00-28EC* body models and *AKG C414 + Shure SM57* mic blend to capture 3–4 round-robin takes of flatpicked single notes (MIDI 38–84), heavy thumb/pick Boom/Alt bass notes (`B` / `A`), palm mutes (`pm`), and bluegrass bass walks.
   - `j45` / `sj200` **Bank (Ample Sound SJ-200 Strummer + Nembrini Guild/Taylor):** Use Ample SJ's Strummer Detect Mode (holding voicings in `C1–B2` while triggering `C4–A5` strum articulations) to capture 3–4 round-robin takes of whole-chord **Downstrokes (`D`)**, **Upstrokes (`U`)**, **Palm-Muted Strums (`P`)**, and **Muted Chucks (`X`)** across all 12 pitch classes for `maj` , `min` , `dom7` , `m7` , `maj7` , `sus2` , `sus4` , `add9` , and `5` chords.
   - `slackkey` **Bank (Nembrini "Weissenborn" + 1936 Martin 0-17):** Capture resonant open-string Taro Patch (`D-G-D-G-B-D`), Wahine, and Mauna Loa samples + natural harmonics (5th, 7th, 12th frets).
   - **10 Nembrini Acoustic Body Impulse Responses (`.wav` IRs):** Load into a WebAudio `ConvolverNode` for a 1-tap **"Make My Electric Sound Acoustic"** switch when playing through your Sonicake Pocket Master or Looper.
2. **Batch 2 — The Electric Guitar Tri-Engine (Real Sampled Electric DI → 8 Neural DSP Archetypes + `.nam` Captures):**
   - **Replace Synthetic `guitar_di.py` Math Waves with Real Multi-Sampled Electric Guitar DI:** Pair a real multi-sampled Electric Guitar DI library on your Mac (such as *Impact Soundworks Shreddage 3 Hydra/Jupiter/Stratus*, *Ample Metal Eclipse / Ample Guitar PF / SC*, or *Orange Tree Evolution Electric*) with your amp plugins so every note has authentic pick-scrape, bridge-flesh palm-mute thump, and magnetic pickup transients.
   - **Headless Rendering Through All 8 Neural DSP Archetypes + `.nam` Captures (`tools/nam_render.py`):**
     - **Archetype: Cory Wong** (`funk` , `popsoul` , `funkpop`): 4th-position Strat compressor clean, 16th-note muted funk scratch chucks (`X`), and triad stabs.
     - **Archetype: John Mayer** (`mayer` , `blues` , `bluesrock`): Scooped tube clean, edge-of-breakup, double-stop 6ths/3rds, and curl bends.
     - **Archetype: Mateus Asato** (`neosoul` , `mayer` , `popsoul`): Boutique shell-voicing clean (`9ths, 11ths, maj7`) and wet singing legato lead.
     - **Archetype: Tim Henson** (`polyphia` , `progclean`): Sparkly compressed clean + piezo blend for hybrid-picked arpeggios, PLUS a rendered **Multivoicer 4-Voice Chord Harmony Bank** for `Jam` and `Beat maker` .
     - **Archetype: Petrucci X** (`progmetal` , `progclean` , `shred`): Soaring liquid lead and stereo doubler rhythm for `Hear it` solo playback.
     - **Archetype: Misha Mansoor** (`progmetal` , `prog`): Precision-Drive ultra-tight low palm-muted chugs (`pm` , `powerpm`), staccato gates, and odd-grouping prog riffs.
     - **Archetype: Gojira X** (`metal`): Crushing heavy rhythm, pick scrapes, and Whammy octave-shift FX samples.

3. ○ ■ **Archetype: Rabea X** (`progclean` , `Beat maker` , `Looper`): **Overlord Dual-Synth & Ambient Pad Drone Bank** into `inst/` so `Jam` can lay warm ambient synth pads under melodic prog solos.
4. **Batch 3 — Baby Audio Humanoid, VocalEssential & Jam Origin MIDI Guitar (All 4 Vocal/Synth Features Uncut):**
   - **1. Tim Henson / Bon Iver "Multivoicer & Humanoid Choir" Instrument (`inst/humanoid-choir/`):** Bounce single vocal/guitar notes through **Baby Audio Humanoid + Tim Henson's Multivoicer** so `Jam` and `Beat maker` play lush, robotic vocoder and synth-choir chords behind your guitar solos.
   - **2. Pre-Built "Vocal Chop" Pad Banks (`inst/vocal-chops/`):** Use **VocalEssential's** `VocalChop` , `VocalTune` , and `VocalVerb` + **Humanoid** to render 2–3 banks of pitch-locked soulful and modern vocal chops ready for Beat Maker's pitch dial and step grid.
   - **3. Hum-to-Tab & Hum-to-Synth in Looper / Riff Scanner:** Pair `Notes from a recording` (Basic Pitch in `Listen`) with your `humanoid` , `Petrucci` , or `Multivoicer` banks so humming a melody into the phone mic immediately plays back as a liquid lead guitar, synth harmony, or vocal chop—and opens directly as a playable guitar tab.
   - **4. Broadcast Spoken Audio Coaching Cues & Count-Ins:** Process count-in tracks (*"One, two, three, four"*) and Strum Machine spoken chord callouts through **VocalEssential's** 18-plugin vocal chain.
   - **5. Jam Origin MIDI Guitar Ground-Truth & Human Groove Templates:** Use Jam Origin on your Mac to generate reference polyphonic MIDI + pitch-bend curves for benchmarking Riff Scanner AND extract real human strumming/fingerpicking micro-timing offsets into `src/audio/strums.ts` .

STAGE 4: Riff Scanner "Three-Sense" Overhaul + Pro Expansions

This stage merges Claude's 6-point plan in `notes/HANDOFF.md` item 2 with our computer-vision fixes and adds five brand-new pro capabilities so Riff Scanner handles any video, Short, audio clip, or live performance:

Part 4A: Fixing Visual Video Tab Scanning (`src/scan/image.ts` & `RiffScanner.tsx`)

1. **Temporal Median Playhead Eraser:** When a moving vertical playhead bar or bouncing ball crosses a fret digit (`9` vs `8` vs `0`), compute the temporal median pixel value across 3–5 scroll-aligned consecutive frames. Because the numbers sit still while the cursor moves, the playhead vanishes mathematically before OCR runs.
2. **Dual-Mode Smooth-Scroll vs. Page-Flip Stitcher:** Automatically classify whether a tab video uses **Smooth Horizontal Scrolling** (tracking vertical bar-line `|` landmarks so repeated identical riffs are never skipped by cross-correlation) or **Static Page-Flip Cards** (detecting abrupt scene cuts inside the tab box, waiting 3 frames for fade-ins to settle, and stacking each card in sequence).

3. **+35% Above-Staff & +20% Below-Staff Crop Headroom:** Expand the vertical crop margin around the 6 staff lines so hammer-on/pull-off arcs (`h, p`), bend labels (`full, 1/2`), vibrato waves (`~`), triplet brackets (`3`), and chord symbols above String 1 are never sliced off.
4. **Audio-Visual Sensor Fusion (Tie-Breaker):** Never skip audio transcription when an on-screen tab is found. Run the audio pitch detector in parallel so whenever visual OCR has low confidence on a blurry digit (`0` vs `8` vs `9`, or `1` vs `7`), the isolated guitar audio's fundamental frequency at that exact timestamp acts as a mathematical tie-breaker, while `Beat This!` locks in the true BPM and rhythm grid.

Part 4B: Fixing Audio-Only Video & Recording Transcription (`transcribe.ts`, `post.ts`, `toTab.ts`)

1. **6-Stem `HT-Demucs` Guitar Isolation + Raw-Mix High-Frequency Transient Cross-Check:** Run `htdemucs_6s_fp16weights.onnx` first to isolate the Guitar stem (unloading Demucs via `MlPlugin.unload` before loading the note and beat ONNX models so Android RAM stays low), and cross-reference note onset times against the unseparated raw mix's high-frequency spectral flux (`2–8 kHz`) so stem separation never softens fast pick attacks.
2. **Tuned Fast-Note Thresholds + Automated "0.5x Slow-Mo Shred Pass":** Lower `minNoteLen` from 11 frames (~128ms) to `3–4` frames (~40ms) with adaptive thresholds (`onset = 0.32, frame = 0.22, amp = 0.10`) in both Basic Pitch and the GAPS Guitar CRNN. Whenever a passage exceeds 130 BPM or has high note density, automatically run a pitch-preserved **0.5x half-speed `SignalsmithStretch` pass** on that audio slice before neural inference, doubling the temporal frame resolution for 16th/32nd runs and soft pull-offs.
3. **True Articulation Tagging (`h`, `p`, `/ \`, `b`, `~`) + Viterbi Same-String Constraint:** Preserve Basic Pitch's 264-bin `contours` matrix / `.pitchBends` and GAPS pitch contours:
   - **Hammer-On (`h`) & Pull-Off (`p`):** Pitch steps by 1–4 semitones while `onsetFlux` shows no fresh pick attack → tag as `h` or `p` and force `src/tabs/fingering.ts` (`planFingers`) to keep both notes on the **exact same string** inside the 4-fret hand box.
   - **Slide (`/ \`):** Continuous smooth pitch slope gliding across ≥2 semitones between notes on the same string.
   - **Bend (`b`) & Vibrato (`~`):** Pitch contour rising ≥0.5–2 semitones and holding/returning within a single note (`b`), or oscillating at 4–8 Hz (`~`).
4. **Dual-Grid Rhythm Quantizer (`16th` vs. `Triplet/Sextuplet` + `32nd Grace Slots`):** Replace the rigid single-note-per-16th-slot quantizer in `notesToTab()` with per-beat likelihood scoring anchored to `Beat This!` downbeats, preserving triplets, fast runs, and grace notes.
5. **"Watch the Guitarist's Hand" Vision Assist:** When scanning a performance video with no on-screen tab, sample 3–4 keyframes of the player's fretting hand so Gemini Vision identifies the **Capo fret** and **Neck Fret Zone** (e.g., *Frets 5–9*), passing that zone as a soft prior into `planFingers()` so the tab lands on the exact same strings and frets used in the video.

Part 4C: 5 New Pro Expansions for Riff Scanner (Added Ideas)

1. **1. Interactive "Uncertain Note" Confidence Glow & 1-Tap Alternate Voicing Cycler:** Highlight any transcribed note or chord where visual/audio confidence was below 80% with a subtle amber ring. Tapping that note pops up a 1-tap radial wheel showing the **2–3 alternative string/fret equivalents** (or neighboring OCR digits) with instant audio preview, recalculating the surrounding 4-fret hand box automatically.
2. **2. Slow-Motion Harmonic Spectrogram Scrubber (*Transcribe!* / *RipX* Style):** Add a toggleable harmonic spectrogram strip directly under the video/audio timeline in Riff Scanner. Dragging your finger slowly across a fast 1-bar lick scrubs the time-stretched audio in place while projecting the live detected pitch and overtone peaks directly onto the Mini Neck card.
3. **3. Automated Tuning & Capo Pre-Classifier (Before Viterbi Fingering):** Before assigning strings and frets, analyze the lowest detected pitches and open-string drone resonances across the whole clip to automatically detect whether the guitarist is in **Standard, Drop D, Half-Step Down (Eb), Taro Patch (Open G), or Capo Fret N**, prompting a 1-tap tuning/capo switch.
4. **4. Spectral Palm-Mute (`PM`) & Natural/Pinch Harmonic (`<12>` / `PH`) Classifier:** Analyze the spectral centroid and high-harmonic decay ratio of each note: rapid high-frequency roll-off with strong low-mid fundamental bump automatically tags **Palm Mutes (`PM`)** on prog/metal chugs, while missing fundamental + intense 2nd/3rd/4th overtone spike tags **Natural or Pinch Harmonics**.
5. **5. Live `noteDetect.ts` Play-Along Upgrades:** Shorten the post-onset lockout from 120–180ms down to **40ms** when the tab expects `h`, `p`, or `/`; widen `expect()` along active bend trajectories; trigger distortion overtone rejection (`-19` / `-28` semitones) only when the lower string has a fresh attack; and relax upper-harmonic inharmonicity tolerances (`h ≥ 5`) on low acoustic wound strings (`MIDI < 55`). Verify all gains with `tools/bench` + Jam Origin MIDI Guitar F1 scores.

STAGE 5: Carbon-Copy "Strum Machine" + Multi-Genre Rhythm Guitar Engine

Build a complete, feature-for-feature carbon copy of **Strum Machine (`strummachine.com`)** wired into `Jam → Play with a band`, `Play` ("Band behind me"), `Make a tab`, and all Floating Cards—expanded so it plays both **realistic acoustic strumming** AND **studio electric rhythm styles**:

- **Real-Instrument Sample Stitching (Zero MIDI Beep Feel):**
  - **Acoustic Mode (Orange Tree + Ample SJ + Nembrini):** Choose *Orange Tree Dreadnought*, *Ample SJ-200*, *Nembrini Weissenborn Slack Key*, or **Dual-Acoustic (Orange Tree Boom-Chick + Bass Runs panned 25% Left, Ample SJ Lush Strums panned 25% Right)**.
  - **Electric Multi-Genre Mode (Electric Tri-Engine):** Use the exact same chord chart engine to stitch real sampled electric guitars rendered through your 8 Neural DSP Archetypes: **Progressive Melodic Metal** (Misha/Petrucci tight palm-muted `pm` chugs, odd-grouping stabs, and Plini/Henson extended `maj7#11/add9` arpeggios), **Cory Wong Funk** (16th-note muted scratch chops and triad stabs), **John Mayer / Mateus Asato Blues & Neo-Soul** (thumb-slap grooves and shell voicings), and **Reggae** (off-beat skanks).

- **Complete Strum Machine "Band Settings" & Automatic Bass Runs:**
  - **2D Strum Feel X/Y Pad:** Blend continuously between *Boom-Heavy vs. Chuck-Heavy* (X-axis) and *Strum-Heavy vs. Cross-Pick/Arpeggio* (Y-axis), plus sliders for **Upstroke Swing**, **Backbeat Emphasis**, and **Dampened Strum Sustain Duration**.
  - **Automatic Diatonic Guitar Bass Runs:** Weave flatpicked acoustic bass walk-ups, walk-downs, and turnarounds between chord changes (`G → F# → Em7`, `C → B → Am7`, diatonic walks into IV and V) with a 3-level frequency toggle (*Off, Occasional, Frequent*).
  - **Voicings, Capo & "Cadd9 Mash":** Select Standard Open Shapes, Virtual Capo, Drop D, Celtic Drop D, Taro Patch (`D-G-D-G-B-D`), and Strum Machine's **"Substitute Cadd9 for C in the Key of G"** toggle.
  - **Standup Bass, Mandolin/Banjo Chop & Wooden Foot Tap Mixer:** Independent volume, pan, octave range (Low/Mid/High/Mix), and pattern controls (2-beat boom-chick, 4-beat walking bass, off-beat mandolin chop, wooden foot tap, or full drum kit).
- **Interactive Cell Chord Grid, Scaler-Style Harmonic Intelligence & YouTube Jam Analyzer:**
  - Subdivided beat cells (2, 3, or 4 chords per bar), per-beat syncopation/pushes, diamond holds, rests, repeat brackets with 1st/2nd endings, Part A / Part B sections, 1-tap **Key Transposition** (all 12 keys), and 1-tap toggle between **Chord Letters (`G - C - D`)**, **Nashville Numbers (`1 - 4 - 5`)**, and **Roman Numerals (`I - IV - V`)**.
  - **Auto-Speedup Trainer:** Automatically increase tempo by `+X BPM` every `N` reps of a song or selected multi-bar loop.
  - **Scaler 2/3-Style "Suggest Next Chord" Palette:** Tapping any cell in the editor pops up voice-led substitutions, borrowed modal chords (Lydian II, Dorian IV, minor iv), and walking-bass passing chords with 1-tap audio preview.
  - **Smart YouTube Backing Track Analyzer:** Paste any YouTube backing track or audio file to auto-detect Key, BPM, and live chord changes, driving the exact same Strum Machine grid and Floating Fretboard Cards.
  - **1-Tap Bridge to `Make a tab` & `Play`:** Tap *"Write a Lead / Flatpicking Break Over This"* on any chart to generate a custom solo or embellished rhythm tab and practice it in `Play` / `Game Mode` while the Strum Machine band backs you up live.

STAGE 6: Beat Maker Groovebox, "Match a Song's Vibe" & "Guitarcatcher"

Following item #6 in `notes/GEMINI-HANDOFF.md`, Claude must **present a visual mockup first for your approval** across Fold cover (`412×915`), Fold inner (`840×757`), and Tablet (`1280×800`), then implement the full hardware-on-glass workstation:

1. **Edge-to-Edge Full-Screen Cockpit (⤢) & Posture-Aware Layouts:**
    - Tapping ⤢ hides all top app banners and navigation bars so the Beat Maker fills 100% of the screen (`st-transport` at pixel 0, with a compact ⤡ exit button).
    - **Unfolded Fold 8 / Tablet:** 4-Quadrant Zero-Scroll Cockpit (Top-Left: 16 Pads with live mini-waveforms & playheads; Top-Right: Step Grid + Drum Layers; Bottom-Left: Keys/Piano Roll + Chord Motion Pills + XY FX; Bottom-Right: 4-Track Looper Rings + Docked Target-Tone Fretboard).
    - **Half-Folded Tabletop Mode:** Step Grid, Looper rings, and Fretboard on the top vertical screen; giant thumb-friendly 16-Pad MPC deck + Footswitch bar on the bottom flat screen.
2. **"🎵 Match a Song's Vibe" (Similar-But-Different Starter Beat Generator):**
    - Upload any song or YouTube link → run `Beat This!`, `soundKey`, and 6-stem `htdemucs_6s` separation to analyze the exact **BPM, time signature, swing %, drum groove DNA** (e.g., *Prog Metal* double-kick & tight chugs vs. *Reggae* one-drop & off-beat skanks vs. *Dylan Gossett Americana* boom-chuck), and **key/chord color**.
    - Automatically build a **similar-but-different starter beat** (mutating secondary syncopations by ~25% with a **"How Close?" slider** and **"🎲 Reroll Variation"** button), generate a cousin chord progression, and offer a 1-tap **"Extract Drum One-Shots from Song"** toggle that slices the cleanest Kick, Snare, and Hi-Hat hits directly from the uploaded song's isolated drum stem onto your pads.
3. **"Guitarcatcher" Live Guitar-to-Synth Mode (<12ms MPE Resynthesis):**
    - Use Stage 3's continuous pitch-contour spine so plucking, bending, and sliding your real guitar strings directly plays Beat Maker polyphonic synths, **Humanoid Choirs**, **Tim Henson Multivoicer harmonies**, **Rabea Overlord synth pads**, 808 bass, or custom sliced pads (`A1–B16`) in real time and records straight into the Piano Roll.
4. **Hardware-on-Glass Groovebox Controls:**
    - **MPC "16 Levels" Mode:** Spread any pad across 16 pads by *Scale-Locked Pitch, Velocity, Filter Warmth,* or *Decay Length*, plus **Tidy Pads** to compact empty pad slots.
    - **Step Grid Direct Touch Gestures:** Drag up/down on a step to set velocity; swipe left/right for `2x/3x/4x` ratchet rolls (`|||`); hold for a radial wheel (`1:4` 4th-bar fill, probability %, pitch lock, and per-row step length for `5/4, 7/8, 6/8` & polyrhythms).
    - **1-Tap Auto-Bass, Kick Pump, 📌 Latched XY FX Trails, 🪄 Auto-Build Song (A–D) & Track Mute Mode.**

STAGE 7: "Pickup Music" Mastery Course Rebuild & Live Phone Stompbox

1. **Rebuild `Learn` into a Pickup Music-Style Deep Mastery System:**
   - Replace surface-level intros with **10 Graded Deep-Dive Mastery Units**: *1. CAGED Fretboard Connection, 2. Close & Spread Triads Across All String Sets, 3. The 7 Diatonic Modes & Modal Mixture, 4. Intervals & Melodic Ear Training, 5. Shell, Drop-2 & Extended Voicings, 6. Chord Voice Leading & Guide Tones, 7. Melodic Improvisation ("Playing the Changes"), 8. Progressive Melodic Metal (Plini / Intervals / Petrucci Style), 9. Dylan Gossett Acoustic Storytelling & Continuous Embellishments,* and *10. Hawaiian Slack Key (Ki Hō'alu) & Taro Patch Mastery*.
   - Structure every day into a **15-Minute Micro-Routine**: *Step 1: Visual Fretboard & Video Breakdown → Step 2: Targeted Motor-Skill Drill (with Auto-Stepping Speed Ramp: +3% after 2 clean passes) → Step 3: Guided Strum Machine / Neural DSP Backing Track Jam → Step 4: Graduation Check-In recorded in Then vs. Now*.
2. **Native Android `Oboe / AAudio` + `MidiPlugin.java` Sub-8ms Live Phone Stompbox:**
   - Build a native Android C++ `Oboe / AAudio` audio plugin and `MidiPlugin.java` (`android.media.midi.MidiManager`) bridge so your Galaxy Fold 8 and Tab S10+ operate as a **<8ms round-trip live guitar multi-FX pedalboard**.
   - Chain live `.nam` amp models, Nembrini acoustic/cab IRs, noise gate, compressor, overdrive, chorus, tempo-synced delay, studio reverb, Guitarcatcher synth, and the 4-track looper—with **automatic bar-synced preset switching** (switching your pedal tone from clean on Verse bars to lead on Solo bars automatically) and two-way USB/BLE MIDI control for your **Sonicake Pocket Master** and **M-Vave Chocolate Plus** 4-button footswitch.
3. **Lightweight Cloud Progress Sync & Google Drive Sound-Pack Streaming:**
   - Sync lightweight JSON state (completed lessons, practice streaks, custom Strum Machine charts, floating card layouts, learned mic overrides) automatically while streaming/caching heavy private Mac plugin sample banks from your personal Google Drive folder into IndexedDB on demand.